# CHAPTER 2

# PROSODIC MARKING IN NOMINAL PREDICATIONS

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## 1. INTRODUCTION

This study aims to indicate that prosody interacts with both the syntactic and the informative makeup of sentences, which gives rise to language specific constraints of the syntaxS-phonology interface. The basic forms that are analyzed in terms of this interface are syntactic constructions which are composed of nominal strings in Turkish, as categorized below:

(i) Modifier-Noun phrases (Adjective+Noun or Noun+Noun phrases),
(ii) Argument-Predicate phrases (with the Arg+Pred order),
(iii) Predicate-Argument phrases (with the Pred+Arg order).

I first exemplify the three types above, then explain the reason why such nominal strings are interesting in Turkish for the present study:

| | | | | | |
|---|---|---|---|---|---|
| (1) | a. | tatlı kedi | | b. | yavru kedi |
| | | cute cat | | | baby cat |
| | | 'cute cat' | | | 'kitten' |
| | | (Mod+N) | | | (Mod+N) |
| | | or | | | or |
| | | 'The cat is cute.' | 'The cat is a kitten.' | | |
| | | (Pred+Arg) | (Pred+Arg) | | |

In (1a), we see a nominal string consisting of an adjective and a noun, which is structurally ambiguous as it can be a noun phrase (NP) with the meaning 'a/the cute cat' or a sentence (S) with the meaning 'The cat is cute'. Likewise, (1b) includes elements from the noun category where the first constituent can be in-

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terpreted as a modifier as seen in the first reading, 'a/the baby cat (kitten)', or it can be interpreted as the predicate of the next element, which leads to a sentential reading, 'The cat is a kitten'. Let us consider (2) now:

| | | | | | | |
|---|---|---|---|---|---|---|
| (2) | a. | kedi | tatlı | b. | kedi | yavru |
| | | cat | cute | | cat | baby |
| | | 'The cat is cute' | | | 'The cat is baby/a kitten' | |
| | | (Arg+Pred) | | | (Arg+Pred) | |

Both (2a) and (2b) involve predication where the first elements are the arguments of the second ones, which constitutes the opposite order of the predicated interpretation in (1).

| | | | | | |
|---|---|---|---|---|---|
| (3) a. | Kaan'ın | kedi[2] | b. | Kedi | Kaan'ın |
| | Kaan-GEN | cat | | cat | Kaan-GEN |
| | 'The cat is Kaan's / | | | 'The cat is Kaan's / | |
| | The cat belongs to Kaan' | | | The cat belongs to Kaan' | |
| | (Pred+Arg) | | | (Pred+Arg) | |

In (3), the expressions are predicative again: The first constituents function as the predicate of the second ones, leading to a predication relation in between. As different from the expressions in (1–3), those in (4), a compound and a possessive phrase, do not involve a predication relation between the nominal elements:

| | | | | | | |
|---|---|---|---|---|---|---|
| (4) | a. | sokak | kedi-si | b. | Kaan'ın | kedi-si |
| | | street | cat-CM | | Kaan-GEN | cat-POSS |
| | | 'a/the street cat' | | | 'Kaan's cat' | |

These examples imply that nominal strings are not ambiguous if they form a compound (4a) or a possessive construction (4b), which are formally distinct but semantically similar. According to Kunduracı (2013, 2017, 2019) expressions such as (4a) are morphological word-formations whereas those such as (4b) are morphosyntactic phrase formations. Despite this formal contrast between a compound and a possessive phrase, their semantics is parallel: Both formations use a functional association rule applying to two arguments (Kunduracı, 2013), as distinct from the rules of predication and modification (Heim & Kratzer, 1998)

2 This expression may also be read as a possessive phrase in which the possessive suffix - *(s)I(n)* is omitted, with the meaning 'Kaan's cat', as can be seen in (4b).

which involve one functor applying to one argument.[3] The current study does not deal with compounds or possessives but with the nominal strings that are ambiguous between a modification and a predication relation such as those in (1–3).

What underlies the multiple interpretations and the potential ambiguities in (1–3) is the intricate category of the nouns and adjectives in Turkish. As highlighted in Göksel (2010), Turkish nominals may be hard to determine categorically: A word can be interpreted as an adjective in a given context, but as a noun, in another. That is, in Turkish (i) many nouns can also be used as modifiers and (ii) some adjectives can be used in noun positions and take the suffixes that normally apply to noun stems. Braun & Haig (2000), Lewis (2000), Özge & Bozşahin (2010), Uygun (2009), and Kunduracı (2013) also address this categorical issue.[4]

For the potential ambiguity in (1–3), the present study suggests a prosodic resolution, pointing to a hierarchical model in agreement with Selkirk (1995), and Gussenhoven (2004), and assumes an extra level in the sentential ones, which is the I-P (intonational phrase, cf. Kan, 2009). Note that this view is different from that of Kabak & Vogel (2001), where the highest level in the prosodic hierarchy is given as the PPh (phonological phrase).

Another emphasis is the informational aspect of the Turkish prosody: I relate the basic notions that are discussed to (contrastive) focus and a constraint on assigning it. This constraint is known "the postverbal (position) constraint" in the Turkish linguistics literature: No stressed-constituent can occur postverbally (e.g. Göksel, 1998: 103). In the current study, I show that this constraint is also valid for structures with nonverbal predicates. This shows that, rather than the lexical category verb, the above constraint is more related to a grammatical function, i.e. the predicate of the structure.

## 2. The Prosodic Resolutions

Turkish regular word-level stress normally falls on the final syllable of a prosodic word (e.g. Göksel & Kerslake, 2005; Erduvanlı Taylan, 2015). In the case of successive affixation, the place of the stress changes accordingly: It falls on the last affix as illustrated below in (5):

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3 There are important formal contrasts between possessive phrases and compounds based on separability, case inflection, omission of *-(s)I(n)*, the obligatory suffixation and the obligatory, paradigmatic nonsuffixation of *-(s)I(n)* in compounds (Kunduracı 2013, 2017, 2019).

4 Kunduracı (2013) proposes a morphological conversion rule, which changes the category of the stem, N→A, in such cases. Her alternative view is to assume that in an NP a noun can modify another noun in a way different from the modification in compounds: In NPs an ascription relation is held between the nouns while in compounds it is an association relation (through an association function).

| | | | | |
|---|---|---|---|---|
| (5) | a. | ke.dí | b. | kedi-lér |
| | | cat | | cat-Pl |
| | | ‘a/the cat’ | | ‘(the) cats’ |
| | c. | kedi-ler-ín | d. | kedi-ler-in-nín |
| | | cat-PL-2.POSS | | cat-PL-2.POSS-GEN |
| | | ‘your cats’ | | ‘of your cats’ |

According to the existing analyses of the Turkish word-level stress, the prosodic word is the locus of the regular word stress in Turkish, which is the rightmost syllable (e.g. Sezer, 1983; Nespor & Vogel, 1986; Kabak & Vogel, 2001).[5] The phonological phrase level stress (PPh) is considered the highest level stress in Kabak & Vogel (2001), and it is assigned to the leftmost element in a PPh, as I exemplify in (6):

| | | | |
|---|---|---|---|
| | ( x | ) | PPh |
| | ( x) | ( x) | PWd |
| (6) | tat.lı | ke.di | |
| | cute | cat | |
| | ‘a/the cute cat’ | | |

The pitch accent assignment rule implies a hierarchical prosodic mechanism: The PPh level stress (the pitch accent) falls on the the last syllable of the leftmost PWd in the PPh, which is *-lı* in *tatlı* above in (6).[6]

Kunduracı (2013) also refers to the PPh stress and shows that the compounds and possessive phrases are parallel phonologically in that both include one more level than the PWd and both take the pitch accent on the final syllable of the left prosodic word. Despite their formal contrasts, i.e. compounds as morphological word-formations vs. possessive phrases as morphosyntactic units, both pattern with NPs in terms of the prosodic hierarchy (Kunduracı, 2013: 16–21). This analysis differs from Kabak & Vogel’s (2001) suggestion that the prosody of com-

[5] However, there are exceptions to regular word stress: (i) many place names such as *Ánkara, Kastámonu, Kónya, Kanáda ‘Canada’*, (ii) some kinship terms such as *ámca* ‘paternal uncle’, and, not surprisingly, (iii) many loan words such as *négatif* ‘negative’, *tiyátro* ‘theatre’. See Sezer (1983), Çakır (2000), Kabak & Vogel (2011), Erguvanlı Taylan (2015).

[6] See Kabak & Vogel (2001: 339), and Kan (2009: 81–82, 97) for the stress in phonological phrases in Turkish.

pounds in Turkish, for which an intermediate Clitic Group (CG) level between the PWd and the PPh is suggested, is different from that of phrases.[7]

As expressed in Kjelgaard & Speer (1999), the initial syntactic structure assigned to an utterance can be determined by its prosodic phonological representation, and from a very early point, speakers use intonation for the syntactic parsing during processing. Kjelgaard and Speer (1999) highlight that in cases when the syntactic parse is wrong, the speaker goes back and uses prosody as a resolver. Hence, in this section we will also have a chance to see if these assumptions are supported with the data from Turkish. In the following subsections, I provide the reader with the pitch tracings and waveforms of a few nominal strings, which were recorded via Sony NWZ-B142 and analyzed on Praat 5.1.44 by myself, as a pilot study.[8]

## 2. 1. A Phrasal Nominal String

Let us begin with the expression in (7), which involves a modification relation between the two nominals:

(7) kadın doktor
woman/female doctor
'a/the female doctor'

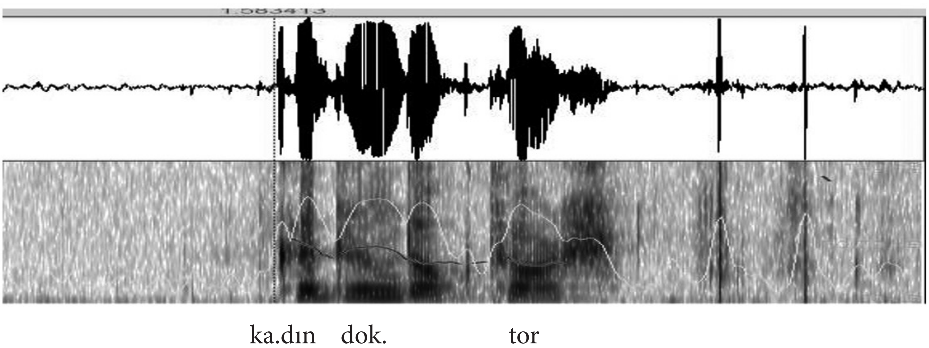



**Figure 1.** Pitch Tracing & Waveform for (7)

In Figure 1, we see a rise in the F0 mapping to the syllable [dɨn] with the highest maximum pitch as 249 Hz, while, for example, the nucleus of the last syllable [tʰoɾ̥] shows 215 Hz. We see perturbations due to obstruents, in this example,

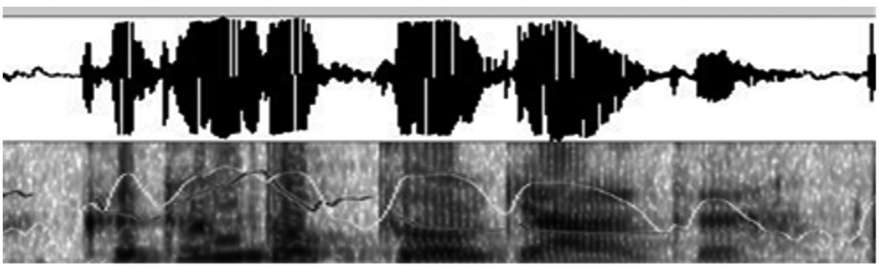
[7] In the current study, I do not assume that the Clitic Group is a level of Turkish phonological structure since it does not differ from the PPh in terms of the main stress of the structure; clitics seem to have no distinctive effect on the prosodic hierarchy. Therefore, I follow Selkirk (1995) and Koch (2008), who suggest to eliminate the CG from the prosodic hierarchy.

[8] The examples in the pilot study were pronounced by a postgraduate native speaker of Turkish.

[$t^h$], in the last syllable. *kadın doktor* with the 'a/the female doctor' interpretation constitutes a PPh with the pitch accent on the rightmost syllable of the leftmost item: [dɨn] in *kadın*.

We will now see the same phrase in a more complex structure, as the subject of a sentence, as introduced in (8):

(8) [Kadın doktor] gel-di
woman/female doctor come-PST/PRF
'A/the female doctor came/has come.'



| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Ka. | | dın | dok.tor | | gel | .di | |
| L | + | H* | | L- | | L-L% | |

**Figure 2.** Pitch Tracing & Waveform for (8)

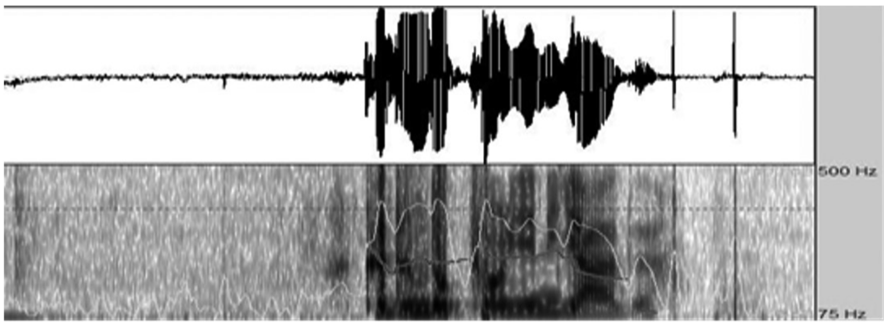
The figure of (8) supports that *kadın doktor* is a PPh since there appears a phrase boundary after the sonorant [ɾ̥] in [$t^h$oɾ̥] before the predicate geldi 'came/has come', which cannot be due to an obstruent's perturbation. There is no boundary between *kadın* and *doktor* since they form a PPh together. This expression is also different from the previous one, (7), in that it includes one more prosodic level, which is the intonational phrase. We can say this seeing the final boundary (tonal marking) which appears as a fall at the end of the utterance. We also see the gradual decline in the F0 towards the end of the sentence, which is a common behaviour of Turkish declaratives in focus-neutral environments. In this respect, I follow Kan's analysis (2009) suggesting that the highest prosodic level in Turkish is the I-P. Now we can see the prosodic structure for (8):

| | | | |
|---|---|---|---|
| ( x | | ) | I-P |
| ( x | ) | ( x) | PPh |
| ( x) | ( x) | ( x) | PWd |
| Ka.dın | dok.tor | gel.di | |

(8') Ka.dın dok.tor gel.di
'The female doctor came/has come.'

Now we consider one more expression, which is similar to but more complex than (8):

| | | | | |
|---|---|---|---|---|
| (9) | Kadın | doktor | yeni | gel-di |
| | woman/female | doctor | just | come-PRF |
| | 'The female doctor has just come.' | | | |




Ka.dın dok.tor ye .ni gel .di
L+ H* H- H* L-L%

**Figure 3.** Pitch Tracing & Waveform for (9)

In this figure, we see a boundary between *doktor* and *yeni*, a phonological phrase boundary, which I show as a rising boundary H- tone. On the other hand, the syllable [ni] in [je.ni] has the highest pitch as 269 Hz, while the [dɨn] syllable shows 246 Hz, which is the pitch accent of the first PPh, and which is lower than the one in the same syllable in Figure 1. This is reasonable as there is another nucleus which is higher than [dɨn] in this expression. What is different in this expression is that we now have two PPhs whose pitch accents are marked as H* above. Since the second one shows higher values, it should be the nuclear accent of the whole utterance as represented in (9'):

```
       (                  x         )          I-P
       (   x        )   ( x         )          PPh
       (   x)   (   x)  ( x ) (   x)           PWd
(9')   Ka.dın  dok.tor   ye.ni  gel-di
       woman/female doctor  just      come-PST/PRF
       'The female doctor has just come.'
```

This expression indicates that when there are more than one PPhs in an I-P, the nuclear accent goes to the right PPh, *yeni geldi* 'has just come' above, which should be another distinctive rule. Accordingly, the proposal in Kan (2009) for the I-P stress placement is supported with the data here: It is assigned to the rightmost PPh in the I-P.

The examples analyzed in this subsection point to the hierarchy in the prosodic structure: Only the accented syllables can take the pitch accent, and only the pitch accented syllables can take the nuclear accent, supporting the proposal in Kratzer & Selkirk (2007).

## 2. 2. A Sentential Nominal String

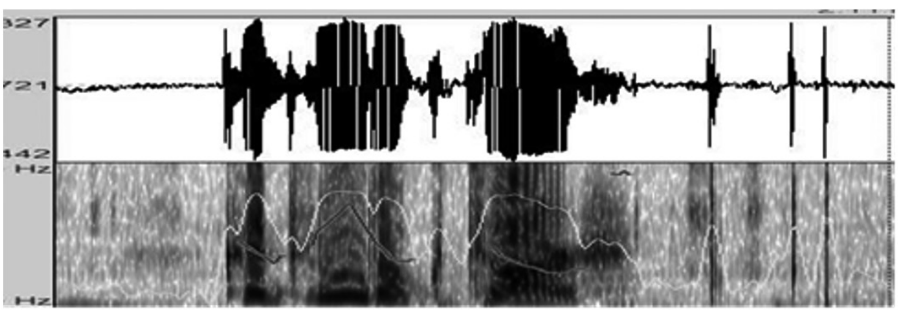
Below is a context where a sentential interpretation of a nominal string takes place easily:

A: Kadın ne iş yap(ı)-yor? 'What does the woman do?

B: (10) Kadın doktor-Ø.

woman/female doctor-AUX

'The woman is a doctor.'



Ka .dın dok .tor

L + H* H*L%

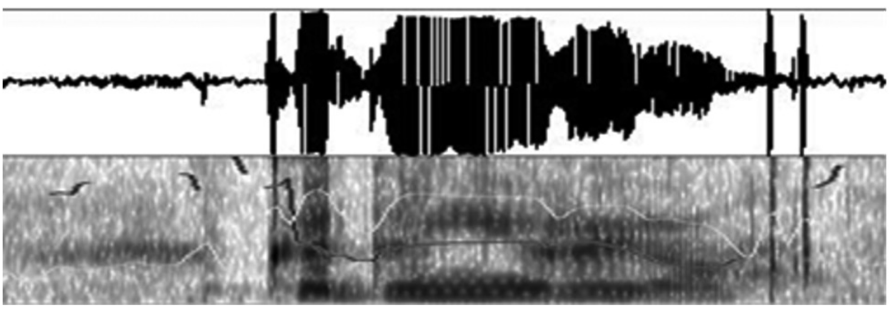
**Figure 4.** Pitch Tracing & Waveform for (10)

This example is interesting in that it differs from the general behaviour of the predicates with a rise on the last syllable, [t$^h$oɾ̥], preceding the I-P boundary L%. This can be regarded as an F0 reset in the second PPh, before which there exists a PPh boundary. The syllable [t$^h$oɾ̥] takes the nuclear accent of the whole unit as 468 Hz, which is higher than the syllable [dɨn], which is 347 Hz. The prosodic structure of the utterance in (10) is represented as follows:

|  |  |  |  |
|---|---|---|---|
|  | ( | x) | I-P |
|  | ( x ) | ( x) | PPh |
|  | ( x ) | ( x) | PWd |
| (10’) | Ka.dın | dok.tor |  |
|  | ‘The woman is a doctor.’ |  |  |

This expression, not only syntactically but also prosodically, signals that the first nominal element does not form a phrase with the second before the S and the I-P levels: The first element, *kadın*, is the subject NP. The nuclear accent is on the rightmost item, *doktor*. The following expression, (11), is presented so as to compare a verbal predicate behaviour and a nominal one like the one in (10). Let us see the context first:

|  |  |  |  |
|---|---|---|---|
| A: |  | Kadın ne yap-ıyor? ‘What is the woman doing?’ |  |
| B: | (11) | Kadın | yürü-yor. |
|  |  | woman/female | walk-PRG |
|  |  | ‘The woman is walking.’ |  |



Ka.dın yü .rü .yor
L + H*H- L+L%

**Figure 5.** Pitch Tracing & Waveform for (11)

In this utterance, there is a verbal predicate *yürüyor* ‘is walking’, which does not take the nuclear accent. Rather, the first word *kadın*, takes the nucler accent although *kadın* is the given information in the context above. While the second syllable of *ka.dın*, [dɨn], shows 256 Hz as the maximum pitch, the second syllable of the predicate *yü.rü.yor*, [ɾy], shows 253 Hz, which is lower than [dɨn]. The values of the progressive suffix *-yor* is not measured because it is one of the exceptional suffixes which are not stressed and which assign the main stress of the word to the immediately preceding syllable. The prosodic structure for this expression can be seen below:

| | | | |
|---|---|---|---|
| | ( x | ) | I-P |
| | ( x) | ( x ) | PPh |
| | ( x) | ( x ) | PWd |
| (11’) | Ka.dın | yü.rü-yor. | |
| | woman/female | walk-PRG | |
| | ‘The woman is walking.’ | | |

The prosodic structure in (11’) is different from the one in (10’) but similar to the one in (8’): The pitch accent is assigned to the leftmost PWd. Although *kadın* is backgrounded in the above context (11), it shows the nuclear pitch, which implies that the backgrounded elements may also be assigned the nuclear pitch (which may imply the role of the prepredicate region prosodically). We might either show the prosodic structure for (11) as above in (11’) or in another way, where the subject NP is not parsed for its own PPh, since backgrounded, which might also account for the reason why the nuclear accent is on the left element of the I-P level. Because there is only one PPh where we get the pitch accent on the leftmost PWd, it may be parsed with the same accent at the I-P level as in (11”) below:

| | | | |
|---|---|---|---|
| | ( x | ) | I-P |
| | ( x | ) | PPh |
| | ( x) | ( x ) | PWd |
| (11”) | Ka.dın | yü.rü-yor. | |
| | woman/female | walk-PRG | |
| | ‘The woman is walking.’ | | |

What can be said through (8–11) is that at the I-P level, the stress is assigned to the rightmost PPh when there are multiple PPhs (9), in the contrary case, i.e. when there is only one PPh, it is assigned to the leftmost element (8, 11) with the exception of nominal predicates (10). According to the prosodic analyses of these expressions, it also appears that to form a phonological phrase in Turkish we need at least two phonological words (9), in contrast to syntactic phrases, which can be established out of single words.

On the other hand, a context where we can get the verbal predicate accented would be when it is focused (contrastively), as illustrated in (12):

A: Kadın hızlı koş(u)-yor. 'The woman is running fast.'

B: (12) Kadın YÜ.RÜ-yor!

woman/female walk-PRG

'The woman is WALKing!

In (12), the nuclear accent is on the predicate (on its second syllable [ɾy]) since the predicate unit is contrastively focused. Now we turn back to the example in (10) *Kadın doktor* 'The woman is a doctor.' and recall that there is an H* on the last syllable [$t^h$oɾ̥] in the nominal predicate. I attribute this case to the syntax and semantics of the predicate. As a nominal predicate, *doktor* is the type <e, t>, which is the same as that of an intransitive verb such as *walk*. Thus, such (stressed) behaviour of the nominal predicate must follow from the lexical category: As a nominal, *doktor* directly defines an entity and its property, which might result in its taking H* regardless of the I-P boundary.[9]

I now turn to another example, where we will see a different sentential interpretation for the string *Kadın doktor,* where both nominal units have definite references this time, as exemplified in the following context:

A: Öğrendiniz mi doktor hangisiymiş?

'Did you find out which one is the doctor?

(13) B: Kadın doktor.

woman/female doctor

'The woman is the doctor.'

Importantly, (13) is different from (10) in that in (13) the predicate *doktor* is also referential, meaning 'the doctor'. In the figure below we see the pitch tracing of this expression:

[9] If one assumes an unheard/unseen copula or auxiliary in (10), the expression in (10) will also parallel the analysis by Göksel (2010), where the presentational focus is realized on the syllable which immediately precedes the auxiliary. This would also show that the immediately preceding elements (syllables in the word, and words in the sentence) are accented, which could help in accounting for the reason why [$t^h$oɾ̥] is stressed before the I-P boundary: the auxiliary effect (if any).

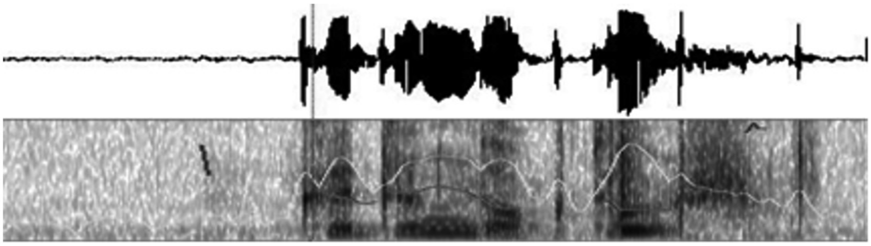



**Figure 6.** Pitch Tracing & Waveform for (13)

This utterance contrasts with (10) in that here we do not see a strong rise but a slight one mapping to the syllable [tʰoɾ̥], the nuclear accent is not assigned here either. The accented syllable is [dɨn] as 264 Hz whereas the maximum pitch for [tʰoɾ̥] is 213 Hz. Thus, the structural ambiguity in referentiality is resolved by the prosody where the information structure must also be at work. For in (10), the subject NP is the theme (given information) while the predicate is the rheme (new information); however, in (13) above, it is vice versa.

Finally, let us underline one more point, in all the nominal constructions given thus far in this section, i.e. the NP (Mod+Head as seen in (8)), the S (Arg+Non-referential Pred as seen in (10)) and the S (Arg+Referential Pred as seen in (13)), we can assign the contrastive focus to any element which takes the pitch accent for informative reasons. However this will cause ambiguity again as there are only two words and very few options for the tonal distinctions. The structural ambiguity due to the informative accentuation is beyond the scope of this study; however, the following expression is presented to illustrate this case:

(14) KADIN doktor

woman/female doctor

a. 'a/the FEMALE doctor'

b. 'The WOMAN is a/the doctor.'

The expression in (14) can mean either 'female doctor, not male' or 'The woman is a/the doctor, not the man.' However, this ambiguity occurs only when the structures are in isolation and not contextualized, of course. In the following section, 2.3, we will see that sentential nominal strings displaying a Pred+Arg relation (rather than the opposite order) are distinctive with regard to the stress assignment place and the focus place as well.

## 2. 3. Another Sentential Nominal String & Postpredicate Position

Turkish displays a free word order for which syntax works in company with the pragmatic information and the discourse felicity (Bozşahin, 2000). Based on certain gapping phenomena, Bozşahin (2000) proposes that the basic word orders are SOV and OSV in Turkish. SVO, OVS, VSO and VOS are possible as well; however, these are restricted in that the postverbal arguments cannot be focused while any words can be focused in the basic, verb-final orders.[10] Erguvanlı Taylan (1984) defines focus as the most informative element in the context and that it is signaled by the word order: The immediate preverbal position is the position for focus while the sentence-initial position is for the topic, and the postverbal element (if any) is for the background information (Erguvanlı Taylan, 1984). On the other hand, Göksel & Özsoy (2000) suggest that there is no designated focus position in Turkish, the sole structural indication of focus is stress: There is no positional restriction except that the focused element is banned from the postverbal region.

In this study, I assume that the analyses which refer to focus, given above, indeed refer to presentational and contrastive/informative focus, respectively. Accordingly, the position for the presentational focus is the immediate preverbal region but the information structure can change it with another apart from the postverbal region. In the present study, I also suggest that the relevant critical impossible position is in fact the postpredicate rather than postverbal as there is also deaccenting in the same position with nominal predicates as well. That is, focus is relevant to a grammatical function, i.e. predicate, rather than a lexical category, i.e. verb.[11]

In (15) below, we see the elements with presentational focus, these are underlined:

| | | | | |
|---|---|---|---|---|
| (15) | a. | Kenzkaan | Talayhan-ı | bul-du. |
| | | Kenzkaan | Talayhan-ACC | find-PST/PRF |
| | | ‘Kenzkaan (has) found Talayhan.’ | | |
| | b. | Talay’ı Kenz buldu. | c. | Kenz buldu Talay’ı. |
| | d. | Talay’ı buldu Kenz. | e. | Buldu Kenz Talay’ı. |
| | f. | Buldu Talay’ı Kenz. | | |

[10] For the word order in Turkish, the reader could also see Enç (1991), Erguvanlı Taylan (1984), and Özsoy (2019), among others.

[11] This study assumes that the grammatical functions such as Pred, Subj, and Obj are arranged by the functional structure of the modular grammar (cf. Bresnan, Asudeh, Toivonen & Wechsler, 2016).

All of the expressions in (15) are perfectly fine in Turkish; the different orders serve for the discourse information as mentioned above; however, an element can also be focused without changing the word order. The focus can be assigned to any preverbal element as well as the verb itself, regardless of the word order, but not to the postverbal element. What the present study highlights, however, is that this constriction also goes for nominal predicates, which is illustrated with the examples below in (16). Unlike the nominal strings analyzed in the sections 2.1 and 2.2, where we can assign the informative focus to each element, those in (16), with the Pred-Arg order, cannot take it on the second element, i.e. the argument:

Pred+Arg Order:

(16) a. KADIN doktor
female/woman doctor
'The doctor is FEMALE/A WOMAN'

b. KAAN'ın kedi
Kaan-GEN cat
'The cat is KAAN's'[12]

a'. *Kadın DOKTOR
'The doctor is female/a woman'

b'.*Kaan'ın KEDİ
'The CAT is Kaan's'

That the postpredicate position cannot be accented, as (16a') and (16b') illustrate, must follow from the fact that these structures are the reversed (or extraposed) versions of the opposite order, Arg+Pred, which is not restricted concerning accenting:

Arg+Pred Order:

(17) a. Doktor KADIN.
'The doctor is FEMALE/A WOMAN'

b. Kedi KAAN'ın.
'The cat is KAAN's'

a'. DOKTOR kadın.
'The DOCTOR is female/a woman.'

b'. KEDİ Kaan'ın.
'The CAT is Kaan's'

When taking account of the expressions in (16) and (17), the parallelism between verbal and nominal predicates, as highlighted in this study, cannot be ignored: The words that follow the predicates cannot be accented in Turkish (probably because those postpredicate elements are necessarily backgrounded). Özge & Bozşahin (2010) refer to the postverbal deaccented position as the "Rightward displacement of the left NP arguments". In their analysis, what is impossible in

[12] (16b) can also be interpreted as a possessive phrase, i.e. 'Kaan's cat', including the omission of the possessive *-(s)I(n)*, which is a distinct structure and not relevant here.

this position might be predication, which is related to rheme, the notion of new information (2010: 151).

Now we will see the pitch tracings of the expressions in (16a, b), respectively:

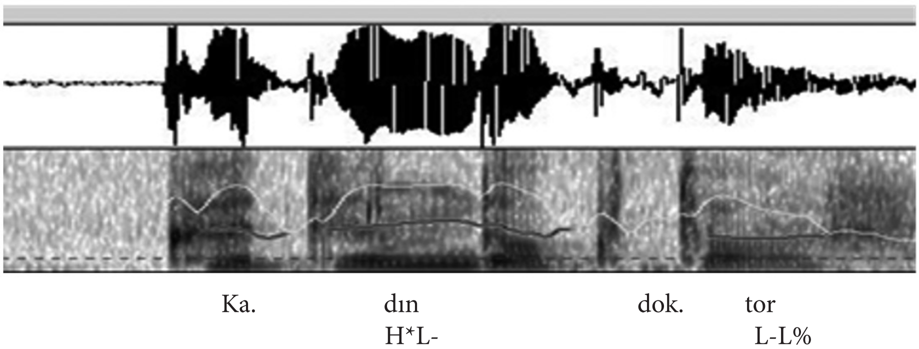



**Figure 7.** Pitch Tracing & Waveform for (16a)

The utterance in (16a) does not indicate a rise mapping onto the word *doktor* unlike (10) and (13) above since *doktor* in (16a) is not the rheme but backgrounded. The syllable [dɨn] has the nuclear accent with the maximum 254 Hz, while [tʰoɾ̥] shows 203 Hz, which is quite lower than that of [tʰoɾ̥] in (10) and (13). We will see the figure for the other extraposed nominal string, i.e. the expression in (16b) below:

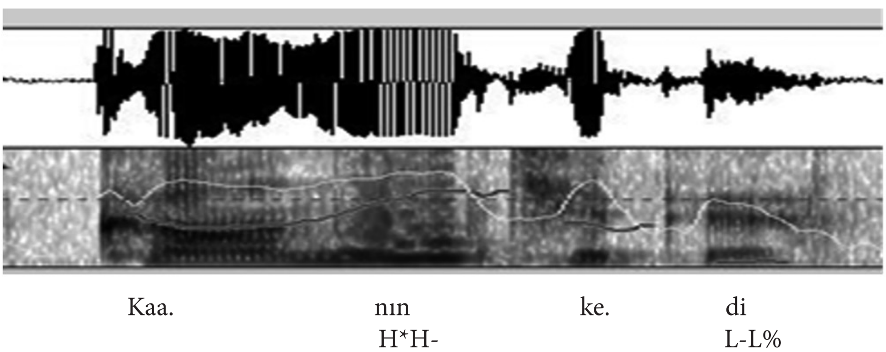



**Figure 8.** Pitch Tracing & Waveform for (16b)

In the sentential *KAAN'ın kedi* expression, it is much easier to see the accentuation in the first element and the deaccentuation in the second. The genitive suffix *-nın* [nɨn] shows the highest maximum pitch, 347 Hz.

In section 2, we considered the prosodic structures in phrasal nominal strings; the structure can be either an NP, where we do not find a phonological phrase boundary in between, or an S with a phonological phrase boundary between the constituents. The second, sentential case can hold either an Arg+Pred or a Pred+Arg order. The first structure is interesting in that the nuclear accent occurs on the the second nominal. This might result from the inherently informative, descriptive characteristics of the nominal predicates. On the other hand, what is common in all these three structures is the pitch accent at the PPh level, which is always assigned to the last syllable of the leftmost PWd in Turkish. The stress in the I-P level, however, shows alteration; according to the findings here, the I-P level stress assignment is related to the number of the PPhs and the type of the predicate. Next, we saw that there is a parallelism between the nominal and the verbal predicates in Turkish: The elements following them are deaccented.

## 3. Conclusions

The current study aimed to show the interaction of syntax and phonology by a few prosodic analyses of the phrasal nominal strings in Turkish. Two previous points have been supported: (i) The phonological parsing needs a hierarchical system and (ii) the top prosodic level, the I-P level, (as assumed here), interacts with the syntax, the information structure and the discourse. Moreover, three findings have been presented: (i) Turkish phonological phrases seem to require at least two phonological words; (ii) unlike the PPh level, the I-P level stress assignment shows alterations based on the number of the PPh, the type of the predicate, and the information structure; (iii) what cannot be accented in Turkish is the postpredicate, rather than only postverbal, position. Future research could investigate the I-P level stress with PPhs which differ in number, and sentences with both nominal and verbal predicates which differ in complexity.

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